JEDNADŽBE DIFERENCIJA

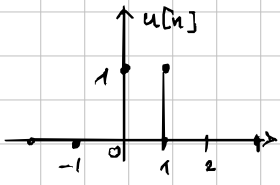
Primjer: Sustav za definiranje jeke

→ miran sustav; poč uvjeti = ∅ $\quad y[-4]=0$ (sve prije polude su poč uvjeti)

$$y[n] = u[n] + \alpha \cdot y[n-N] \qquad u[n] = \begin{cases} 1, & n=0,1 \\ 0, & \text{„inače} \end{cases}$$

$$y[n] = \underbrace{u[n]}_{\text{ovisi o ulazu u sadašnjosti}} + \underbrace{0{,}6\, y[n-4]}_{\text{ovisi o izlazu u prošlosti}}$$

MEMORIJSKI!



$$y[0] = u[0] + 0{,}6 \cdot \underbrace{\cancel{y[-4]}}_{\text{poč uvjet} = \emptyset} \Rightarrow \boxed{y[0]=1}$$

$$y[1] = u[1] + 0.6 \cdot y[-3] = 1 \Rightarrow \boxed{y[1]=1}$$

$$y[2] = u[2] + 0{,}6 \cdot y[-3] = 0 \qquad \boxed{y[2]=0}$$

$$\dots \boxed{y[3]=0}$$

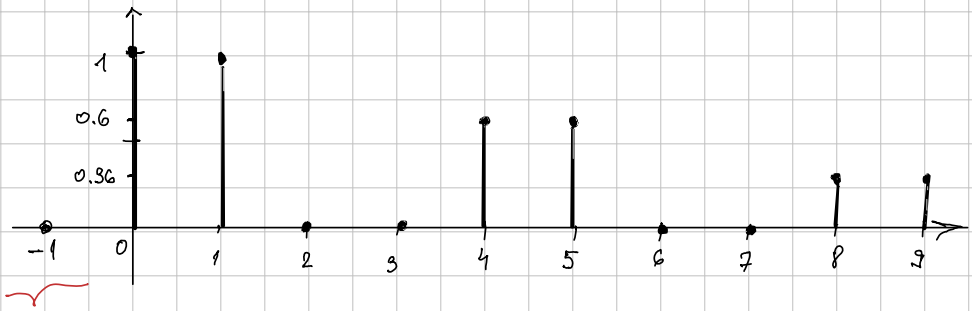
$$y[4] = \underset{\overset{\shortparallel}{0}}{u[4]} + 0.6 \cdot y[0] \Longrightarrow \boxed{y[4]=0.6}$$

$$y[5] = 0 + 0.6 \cdot y[1] \qquad \boxed{y[5]=0.6}$$

$$y[6] = 0 + 0.6 \cdot y[2] \qquad \boxed{y[6]=0}$$

$$\dots \boxed{y[7]=0}$$

$$y[8] = \overset{\overset{0}{\shortparallel}}{u[8]} + 0.6 \cdot y[4] \Rightarrow \boxed{y[8] = 0.6^2} = 0.36$$



prije nule su zadani poč. uvjeti za koje smo rekli da su ∅

→ budući da je $N=4$ → trebali smo 4 poč. uvjeta

→ kažemo da je jednadžba 4. kog reda

opći oblik jed. diferencija:

$$y[n] + a_1 y[n-1] + a_2 y[n-2] + \dots + a_N \cdot y(n-N)$$
$$= b_0 \cdot u[n] + b_1 u[n-1] + \dots + b_N u[n-N]$$

*koliko zakasnimo ulaz toliko je i izlaz

$$\Rightarrow \boxed{y[n] + \sum_{m=1}^{N} a_m \cdot y[n-m] = \sum_{m=0}^{N} b_m \cdot u[n-m]}$$

*„padajuća" opcija: $y[n+N] + a_1 y[n+N-1] + \dots + a_N y[n] =$
$b_0 u[n+N] + b_1 u[n+N-1] + \dots + b_N u[n]$

Primjer: $y[n+2] + 0.5 y[n+1] + 0.06 \cdot y[n] = u[n+1]$

$n=0 \longrightarrow y[2] + 0.5 y[1] + 0.06 y[0] = u[1] \longrightarrow n \geq 2$

da bi njega mogli izračunati trebaju nam biti zadani poč. uvjeti $y[1]$ i $y[0]$ (jer su prje)

npr zadano: $y[1] = -1$
$y[0] = 1$

$u[n] = 0.5^n \mu[n]$

• pomaknemo jednadžbu: $y[n+1] + 0.5 y[n] + 0.06 y[n-1] = u[n]$

$n=0 \longrightarrow y[1] + 0.5 y[0] + 0.06 y[-1] = u[0] \longrightarrow n \geq 1$

poč uvjeti

$y[0] = 1$

$y[-1] = ?$ uvrstimo da dobijemo

$-1 + 0{,}5 \cdot 1 + 0{,}06 \cdot y[-1] = 0.5^0$

$0.5 + 0.06 y[-1] = 2 \longrightarrow \boxed{y[-1] = 25}$

pomaknemo opet: $y[n] + 0.5 y[n-1] + 0{,}06 y[n-2] = u[n-1]$

$n=0 \rightarrow y[0] + 0.5 y[-1] + 0.06 y[-2] = u[-1] \longrightarrow n \geq 0$

poč uvjeti

*dobijemo zadano

$y[-1] = 25$

$y[-2] = ?$

$1 + 0.5 \cdot 25 + 0.06 \cdot y[-2] = \underbrace{0.5^{-1} \cdot \mu[-1]}_{0} \longrightarrow \boxed{y[-2] = -225}$

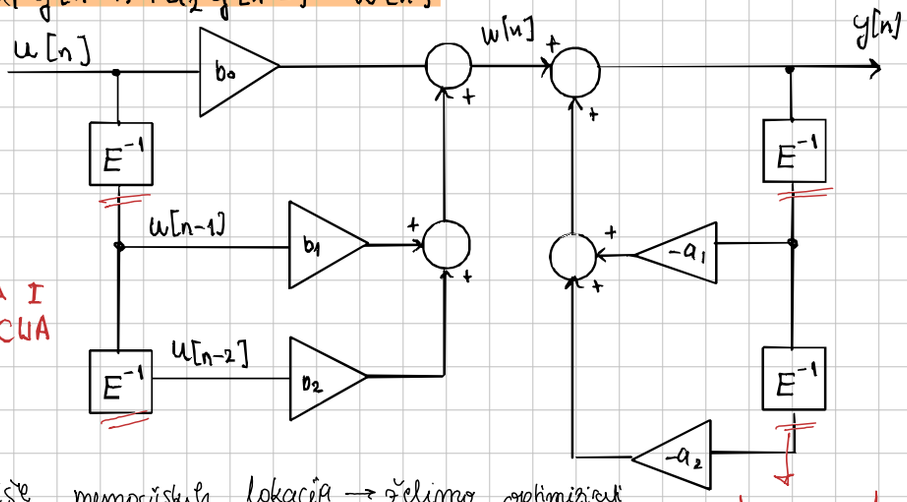
⇝ Direktna I realizacija

$$y[n] + \sum_{m=1}^{N} a_m y[n-m] = \sum_{m=0}^{N} b_m u[n-m]$$

$$y[n] + a_1 y[n-1] + a_2 y[n-2] = b_0 u[n] + b_1 u[n-1] + b_2 u[n-2]$$

$$b_0 u[n] + b_1 u[n-1] + b_2 u[n-2] = w[n]$$

$$y[n] + a_1 y[n-1] + a_2 y[n-2] = w[n]$$



DIREKTNA I REALIZACIJA

malo previše memorijskih lokacija → želimo optimizirati

↳ memorijske lokacije

Direktna II realizacija

1. jednadžba

$$y[n] + a_1 y[n-1] + a_2 \cdot y[n-2] = u[n]$$

↳ samo jedan ulaz umjesto izvora više pomaknutih

rezultat tog ulaza je neki $y_a[n]$ → pomoćna jednadžba

$$y_a[n] + a_1 y_a[n-1] + a_2 y_a[n-2] = u[n]$$

$u[n] \longrightarrow y_a[n]$

$u[n-1] \longrightarrow y_a[n-1]$

$u[n-2] \longrightarrow y_a[n-2]$

vremenska nepromjenivost

$b_0 \cdot u[n] \longrightarrow b_0 \cdot y_a[n]$

$b_1 \cdot u[n-1] \longrightarrow b_1 \cdot y_a[n-1]$

homogenost

$b_2 \cdot u[n-2] \longrightarrow b_2 \cdot y_a[n-2]$

linearnost

$\boxed{1.}$ napisali smo pomoćnu jednadžbu koja kaže da je na ulazu samo $u[n]$ → nij sikroz isto kao zadanu

$\boxed{2.}$ izlaz pomoćne = $y_a[n]$

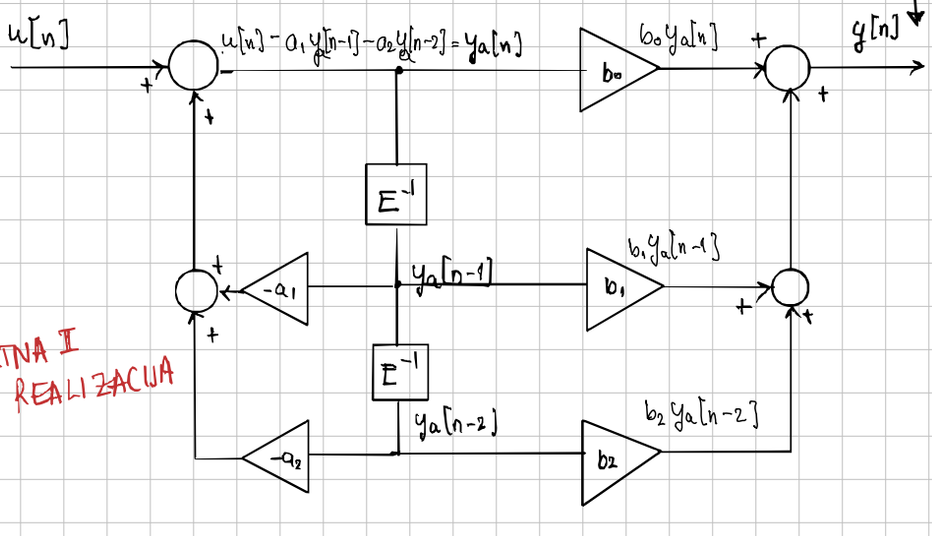
$\boxed{3.}$ Zbog toga zbroj takvih pomaknutih y_a daju izlazni sustav

2. jednadžba

$$b_0 \cdot u[n] + b_1 \cdot u[n-1] + b_2 u[n-2] \longrightarrow \underbrace{b_0 y_a[n] + b_1 y_a[n-1] - b_2 y_a[n-2]}_{y[n]}$$

↓ izlaz 1. sustava!

$$y_a[n] = u[n] - a_1 \cdot y_a[n-1] - a_2 y_a[n-2]$$



DIREKTNA II REALIZACIJA

Zapis jednadžbe diferencija pomoću operatora kašnjenja

$$y[n] + a_1 y[n-1] + \dots + a_N y[n-N] = b_0 u[n] + b_1 u[n-1] + \dots + b_N u[n-N]$$

$$E^{-1} u[n] = u[n-1] \dashrightarrow E^{-k} u[n] = u[n-k]$$

$$1 \cdot y[n] + a_1 \cdot E^{-1} \cdot y[n] + \dots + a_N \cdot E^{-N} y[n] \Longrightarrow y[n]\overbrace{\left(1 + a_1 E^{-1} + \dots + a_N E^{-N}\right)}^{A(E^{-1})}$$

nema kašnjenja (E^0)

$$= b_0 \cdot u[n] + b_1 \cdot E^{-1} u[n] + \dots + b_N E^{-N} u[n]$$

$$\longrightarrow u[n]\underbrace{\left(b_0 + b_1 E^{-1} + \dots + b_N \cdot E^{-N}\right)}_{B(E^{-1})}$$

$$\rightsquigarrow \boxed{A(E^{-1}) y[n] = B(E^{-1}) \cdot u[n]}$$

ukupni odziv $y[n] = y_h[n] + y_p[n]$

HOMOGENO $\Rightarrow u[n]$ — PARTIKULARNO

$A(E^{-1}) y_h[n] = 0$ — nema veze sa ulazima, samo sa poč. uvj.

$A(E^{-1}) y[n] = B(E^{-1}) u[n]$

$$y[n] \Rightarrow A(E^{-1}) y_h[n] + A(E^{-1}) y[n] = B(E^{-1}) u[n]$$

$$y[n] \Longrightarrow A(E^{-1})\left(y_h[n] + y_p[n]\right) = B(E^{-1}) u[n]$$

// dokazali da je ukupni odziv zbroj homogenog i partikularnog

Primjer: $y[n] - 0.9 y[n-1] + 0.2 y[n-2] = u[n] + u[n-1] + u[n-2]$

Zadano: $u[n] = -9(-0.9)^n \mu[n]$

$y[-1] = 6 \qquad y[-2] = -3$

* često će nam biti rečeno da su sustavi kauzalni i da nam je i pobuda kauzalna

kauzalnost sustava $\neq$ kauzalnost signala!

Korak 1: riješi homogenu jednadžbu

$$y[n] - 0.9 y[n-1] + 0.2 y[n-2] = 0$$

oblik hom rj. za diskretne sustave $y_h[n] = C \cdot q^n$

$$C q^n - 0.9 C q^{n-1} + 0.2 C q^{n-2} = 0$$

$$C q^{n-2}\left(\underbrace{q^2 - 0{,}9 q^1 + 0.2}_{\text{mora biti } = 0!}\right) = 0 \qquad C \neq 0,\ q \neq 0$$

$\Longrightarrow$ KARAKTERISTIČNI POLINOM $\quad q^2 - 0{,}9 q + 0.2 = 0$

karakt. vrij. / karakt. frekv. ili vlastite vrijedn. / vlastite frekv.

$$q^2 - 0.5q - 0.4q + 0.2 = 0$$

$$q(q - 0.5) - 0.4(q - 0.5) = 0 \rightarrow (q - 0.5)(q - 0.4) = 0$$

karakteristične vrijednosti ili frekvencije $\boxed{q_1 = 0.5 \qquad q_2 = 0.4}$

$$\boxed{y_h[n] = C_1 \cdot 0.5^n + C_2 \cdot 0.4^n}$$

* da smo dobili 1 ili više istih q: $q_1 = q_2 = 0.5$ $\quad q_3 = 0.4$

$$y_h[n] = C_1 \cdot 0.5^n + C_2 \cdot n \cdot 0{,}5^n + C_3 \cdot 0{,}4^n$$

kratnost 2 $\hookrightarrow n^{k-1}$ je dodani član

* da smo dobili $q_1 = 0.3 + 0.4j$, $q_2 = 0.3 - 0.4j$ } sigurno dolaze u konjugiranom kompleksu, provjeriti.

$y_h[n] = C_1 \cdot (0{,}3 + 0.4j)^n + C_2 (0.3 - 0.4j)^n$ $\longrightarrow$ ali bilo bi dobro da preko cos i sin!

$$\dots\ y_h[n] = C_1 \left(0.5 \cdot e^{-j0.93}\right)^n + C_2 \left(0.5 \cdot e^{j93}\right)^n$$

$$= C_1 \left(0.5^n \cdot e^{-j0.93n}\right) + C_2 \left(0.5^n \cdot e^{j93n}\right)$$

$C_1 = \frac{C}{2} e^{-j\varphi}$ $\qquad C_2 = \frac{C}{2} e^{j\varphi}$

$$\rightarrow y_h[n] = \frac{C}{2} e^{-j\varphi} \cdot 0.5^n \cdot e^{-j0.93n} + \frac{C}{2} e^{j\varphi} \cdot 0{,}5^n \cdot e^{j0.93n}$$

$$= \frac{C}{2} 0.5^n \left[e^{-j(0.93 + \varphi)} + e^{j(0.93 + \varphi)} \right]$$

$$= \frac{C}{\cancel{2}} \cdot 0{,}5^n \cdot \cancel{2} \cos(0.93 + \varphi) \longrightarrow \boxed{y_h[n] = C \cdot 0{,}5^n \cdot \cos(0.93 + \varphi)}$$

korak 2.: riješi partikularno rješenje

→ rj koje ovisi o pobudi! $u[n] = -9 \cdot (-0.9)^n \mu[n]$

* partikularno rj je isto rj koje je oblika i pobuda

$$y_p[n] = K \cdot (-0.9)^n \quad , za\ n \geq 0$$

* tek na kraju množimo sa stepom radi jednostavnosti

$$\Rightarrow y[n] - 0.9y[n-1] + 0.2y[n-2] = u[n] + u[n-1] + u[n-2]$$

↳ uvrstimo y_p

$$K \cdot (-0.9)^n - 0.9 \cdot K \cdot (-0.9)^{n-1} + 0.2 \cdot K \cdot (-0.9)^{(n-2)} = -9 \cdot (-0.9)^n - 9 \cdot (-0.9)^{n-1} - 9 \cdot (-0.9)^{n-2}$$

$$K - 0.9K \cdot (-0.9)^{-1} + 0.2 \cdot K \cdot (-0.9)^{-2} = -9 - 9 \cdot (-0.9)^{-1} - 9 \cdot (-0.9)^{-2}$$

$$K\left[1 + \cancel{0.9} \cdot \frac{-1}{\cancel{0.9}} + 0.2 \cdot (-0.9)^2\right] = -9\left[1 + (-0.9)^{-1} + (-0.9)^{-2}\right]$$

$$\rightsquigarrow \boxed{K = -4.5} \longrightarrow \boxed{y_p[n] = -4{,}5 \cdot (-0{,}9)^n \quad za\ n \geq 0}$$

* moguće pobude i partikularna:

$u[n]$		$y_p[n]$
A		K
$A \cdot r^n$	$r \neq g_i$	$K \cdot r^n$
$A \cdot r^n$	$r = g_i$	$K \cdot n \cdot r^n$
polinomu → $A \cdot n^M$		$K_0 + K_1 n + K_2 n^2 + \dots + K_M n^M$
$T^N \cdot n^M$		$r^N (K_0 + K_1 n + \dots + K_M \cdot n^M)$
$A \cos(w_0 n)$		$K_1 \cdot \cos(w_0 n) + K_2 \cdot \sin(w_0 \cdot n)$
$B \sin(w_0 n)$		

→ za kombinacije ovih možemo odvojeno računati $y_{p1}[n] + y_{p2}[n] + \dots = y_p[n]$ (vrijedi linearnost)

$u[n] = -9(-0{,}9)^n \mu[n]$

$y[-1] = 6$

$y[-2] = -3$

poč. uvjet

Ukupno rješenje: $y[n] = y_h[n] + y_p[n]$

$$y[n] = C_1 \cdot 0{,}5^n + C_2 \cdot 0{,}4^n - 4{,}5 \cdot (-0{,}9)^n$$

* odziv vrijedi NAKON POBUDE → ne možemo ravno uvrstiti

→ „premotati" $y[-1]$ i $y[-2]$ u $[0]$ i $[1]$

$$y[n] - 0.9y[n-1] + 0.2y[n-2] = u[n] + u[n-1] + u[n-2]$$

$$y[0] - 0.9y[-1] + 0.2y[-2] = u[0] + u[-1] + u[-2]$$

$$y[0] = u[0] + u[-1] + u[-2] + 0.9y[-1] - 0.2y[-2]$$

$$y[0] = -9 \cdot (-0{,}9)^0 + \emptyset + \emptyset + 0.9 \cdot 6 - 0{,}2 \cdot (-3) \Rightarrow \boxed{y[0] = -3}$$

$$y[1] = u[1] + u[0] + u[-1] + 0{,}9\,y[0] - 0.2\,y[-1]$$

$$y[1] = -9 \cdot (-0{,}9)^1 - 9 \cdot (-0{,}9)^0 + 0 + 0{,}9 \cdot (-3) - 0{,}2 \cdot 6 \Rightarrow \boxed{y[1] = -4{,}8}$$

to se početni uvjeti

uvrštavamo u odziv

$$y[0] = C_1 \cdot 0{,}5^0 + C_2 \cdot 0{,}4^0 - 4{,}5(-0{,}9)^0 = -3 \rightarrow C_1 + C_2 = -1.5$$

$$C_1 = -1.5 - C_2$$

$$y[1] = C_1 \cdot 0{,}5 + C_2 \cdot 0{,}4 - 4{,}5 \cdot (-0{,}9) = -4{,}8$$

$$\Rightarrow \boxed{C_1 = -94.5 \quad C_2 = 96}$$

$$\rightsquigarrow \boxed{y[n] = \underbrace{\overbrace{-94.5 \cdot 0{,}5^n + 96 \cdot 0{,}4^n}^{\text{iz } y_h[n]}}_{\text{prirodni / prijelazni odziv}} \underbrace{\overbrace{-4.5 \cdot (-0{,}9^n)}^{\text{iz } y_p}}_{\text{prisilni odziv}}} \cdot \mu[n] \text{ jer } n \geq 0!$$

NEPOBUĐENI ODZIV

$y[-1] = 6 \qquad y[-2] = -3 \longrightarrow$ imamo samo homogeno rješenje

$y_h[n] = C_1 \cdot 0{,}5^n + C_2 \cdot 0{,}4^n$ ↳ da bi dobili nepobuđeni odziv, direktno uvrštavamo te y u izraz homogenog rješenja

$y_h[-1] = C_1 \cdot 0{,}5^{-1} + C_2 \cdot 0{,}4^{-1} = 6$

$y_h[-2] = C_1 \cdot 0{,}5^{-2} + C_2 \cdot 0{,}4^{-2} = -3$

$C_1 = 18 \qquad C_2 = -12$

$\Rightarrow \boxed{y_0[n] = 18 \cdot 0{,}5^n - 12 \cdot 0{,}4^n}$

MIRNI SUSTAV

$y[-1] = 0 \qquad y[-2] = 0 \qquad u[n] = -9 \cdot (-0{,}9)^n \mu[n]$

poč. uvjeti su 0; prije pobude (znači ima pobudu)

$y_m[n] = C_1 \cdot 0{,}5^n + C_2 \cdot 0{,}4^n - 4{,}5 \cdot (-0{,}9)^n$ * kompletno rješenje

*moramo preračunati $y[-1]$ i $y[-2]$ jer su bili prije pobude

$y[0] = u[0] + u[-1] + u[-2] + 0{,}9\, y[-1] - 0{,}2\, y[-2]$

$= -9 + \emptyset + \emptyset + \emptyset + \emptyset \longrightarrow \boxed{y[0] = -9}$

$y[1] = u[1] + u[0] + \emptyset + 0{,}9\, y[0] - \emptyset$

$= -9 \cdot (-0{,}9) - 9 + 0{,}9 \cdot (-9) \longrightarrow \boxed{y[1] = -9}$

$y_m[0] = C_1 \cdot 0{,}5^0 + C_2 \cdot 0{,}4^0 - 4{,}5 \cdot (-0{,}9)^0 = -9$

$y_m[1] = C_1 \cdot 0{,}5^1 + C_2 \cdot 0{,}4^1 - 4{,}5 (-0{,}9)^1 = -9$

$C_1 = -112{,}5 \qquad C_2 = 108$

$\rightsquigarrow \boxed{y_m[n] = (-112{,}5 \cdot 0{,}5^n + 108 \cdot 0{,}4^n - 4{,}5 \cdot (-0{,}9)^n)}\, \mu[n]; \ (n \geq 0)$

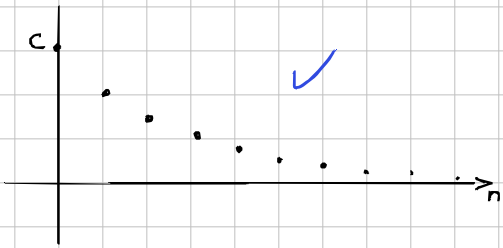
$\Rightarrow$ kada zbrojimo to dvoje = TOTALNI ODZIV

$y[n] = \underbrace{18 \cdot 0{,}5^n - 12 \cdot 0{,}4^n}_{y_0} \underbrace{- 112{,}5 \cdot 0{,}5^n + 108 \cdot 0{,}4^n - 4{,}5 \cdot (-0{,}9)^n}_{y_m}$

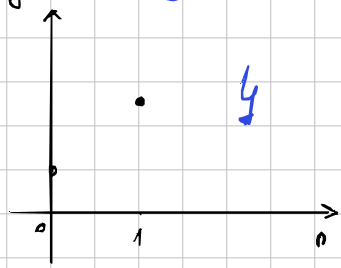
$y[n] = -94{,}5^n + 96 \cdot 0{,}4^n - 4{,}5 \cdot (-0{,}9)^n$

Nepobuđeni odziv u ovisnosti frekv. (oni g)

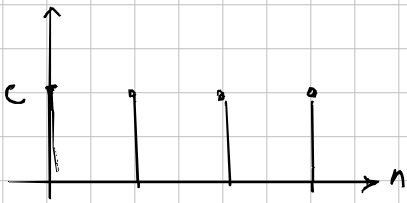
$y_0[n] = c \cdot 0{,}5^n$



$y_0[n] = c \cdot 2^n$



$y_0[n] = c \cdot 1^n$



↳ marginalna stabilnost

→ kakve god pobude imali, ako zbrojimo jedan rastući nepobuđeni odziv sa pobudom, naš odziv će stalno rasti

BIBO: ako na ograničenu pobudu odziv nije ograničen → sustav NIJE STABILAN

↓

SUSTAV U OVAKVOM SLUČAJU NIJE STABILAN

→ direktno iz nepobuđenog odziva vidimo hoće li biti stabilan

SUSTAV (BIBO) STABILAN: ako su mu karakteristične frekv. manje od 1! (po mnogoj aps. v.)

$|g| < 1$ stabilno

$|g| > 1$ nestabilno

$|g| = 1$ granično stabilno

$|g_1| = |g_2| = 1$ nestabilno

$*\, g_1 = g_2$ → kćino isk